INCH-POUND

MIL-V-6618C
8 JANUARY 1993
SUPERSEDING
MIL-V-6618B
4 FEBRUARY 1980

MILITARY SPECIFICATION

VALVE, STOP CHECK, PURGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a purge valve for use in an aircraft manifold pressure gage line.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage.
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-970	Standards & Specifications, Order Of Preference For The Selection Of
MIL-STD-2073-1	DOD Material Procedures For Development & Application Of Packaging Requirements
AN815	Union & Flared Tube
MS33649	Boss, Fluid Connection - Internal-Straight Thread

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), a sample shall be subject to first article inspection (see 6.3) in accordance with 4.2.

3.1.1 Components. The purge valve shall consist of a valve, button, mounting nuts, and lockwashers.

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3.2 Materials.

3.2.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the valve, except where magnetic materials are essential.

3.2.1.1 Recovered materials. Recovered materials are used to the maximum extent possible without jeopardizing the end use of the item.

3.2.2 Metals. Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during normal service life.

3.2.3 Protective treatment. When materials are used in the construction of the valve that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.4 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-970 except as provided in the following paragraph.

3.2.5 Standard parts. Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and , if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction. The valve shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

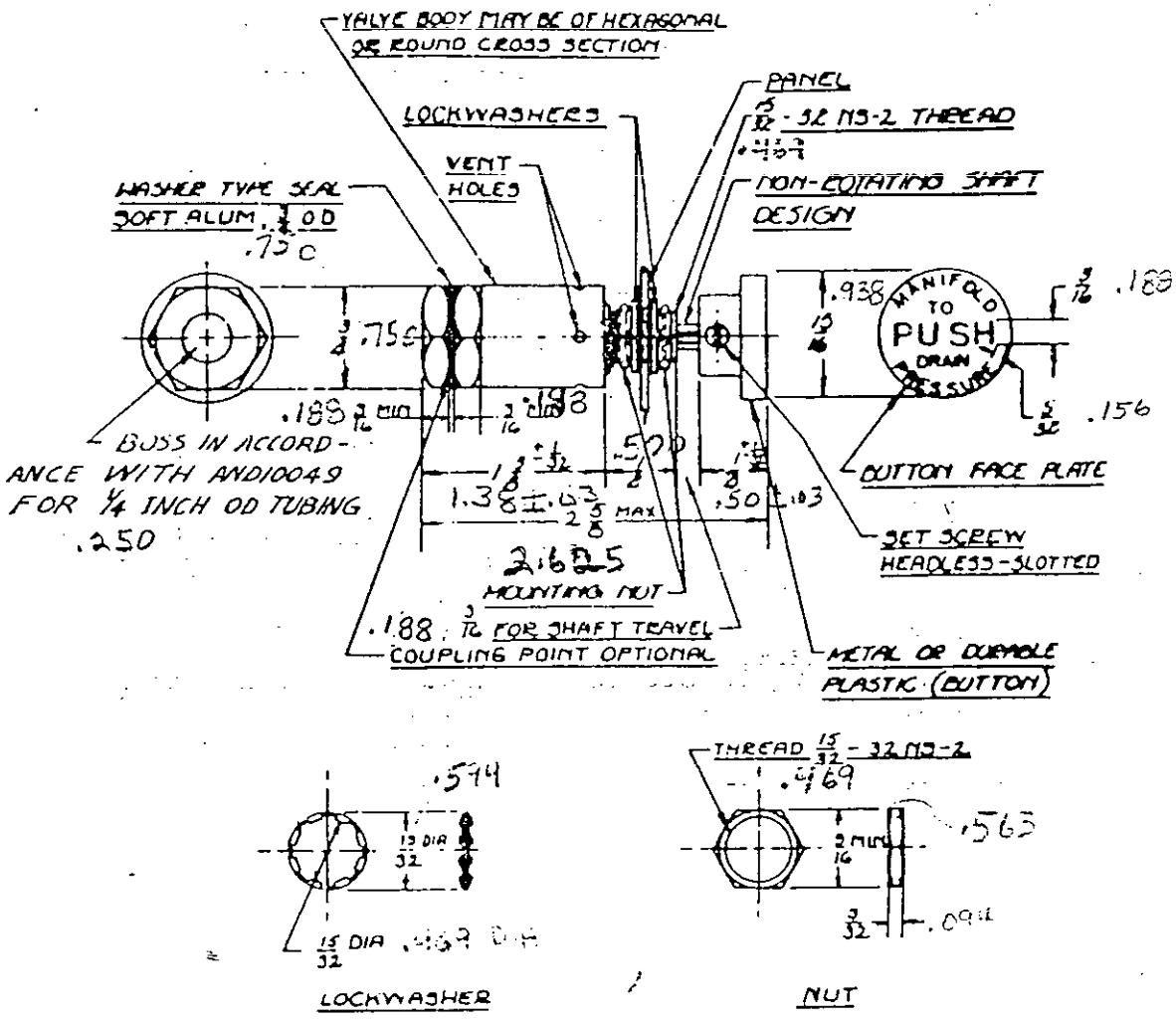
3.3.1 Bosses and ports. The back of the valve shall be provided with a boss in accordance with Drawing MS33649 for 0.25 OD tubing. Venting ports shall be provided on the sides of the valve and of such area as specified herein for flow.

3.3.2 Plugs. The threaded boss shall be provided with a plug of acceptable material and design.

3.3.3 Valve mechanism.

3.3.3.1 Normal position. The valve mechanism shall normally be held in a closed position by a spring, and shall be so constructed that additional air pressure applied at the rear fitting will tend to seal the valve more firmly.

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NOTES:

1. Dimensions are in inches.
2. Tolerance: Decimals ± .015 unless otherwise specified.

FIGURE 1. Valve

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3.3.3.2 Secondary position. The secondary or vent position shall be attained by pressing the valve shaft button thereby relieving the valve seat.

3.3.3.3 Venting. The venting shall be accomplished through holes drilled in the sides toward the front of the valve as shown on figure 1.

3.3.3.4 Clearance. The clearance dimension for air flow through the valve shall be not less than the equivalent of a standard AN815-4 union, and the spring shall not compress to the point where it will restrict the flow at full shaft travel or wide open condition.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-STD-100.

3.5 Button faceplate. An engraved or permanently marked button conforming to figure 1 shall be provided separately with each valve. The button shall be finished in durable dull black, except for the markings on the face plate of the push button.

3.5.1 Faceplate markings. The faceplate marking shall conform to figure 1 and shall be finished in durable dull-white or natural-white metal finish with letters as shown on figure 1. The markings shall be durable to withstand usage encountered in service.

3.6 Mounting nuts and lockwashers. Two mounting nuts and two lockwashers, as shown on figure 1, shall be furnished separately with each valve.

3.7 Weight. The weight of the complete valve shall not exceed 0.10 pound.

3.8 Dimensions. The dimensions shall be as shown on figure 1.

3.9 Screw threads. Unless otherwise specified, machine screw threads shall conform to MIL-S-7742.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 Use of AN or MIL designations. AN or MIL designations shall not be applied to a product, except for first article test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for first article testing.

3.11 Workmanship.

3.11.1 General. The valve, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges, accuracy of dimensions, marking of parts and assemblies, alignment of parts, and tightness of assembly screws and bolts.

3.11.2 Cleaning. The valve shall be thoroughly cleaned, and metal chips and other foreign material removed during and after final assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).

4.3 First article inspection.

4.3.1 Sampling instructions. First article test samples shall consist of four valves. Samples shall be identified as required and forwarded to the activity responsible for testing, designated in the authorization from the responsible activity (see 6.3).

4.3.2 Tests. First articles shall be performed on preproduction samples consisting of all examinations and tests specified herein. Offering vendor/ contractor shall at no cost to the government, issue certifications, drawings, and test reports demonstrating that parts meet specification requirements. First article shall be approved prior to production of valves.

4.4 Quality conformance inspection. The quality conformance inspection test shall consist of individual tests, sampling tests, and special test, as specified in this specification.

4.4.1 Individual tests. Each valve shall be subjected to the following tests. In addition, each valve shall be subjected to any other tests specified herein which the inspector considers necessary to determine compliance with the requirements of this specification.

4.4.1.1 Examination of product. Each valve shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

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4.4.1.2 Leak test. The valve shall be connected to a mercury manometer and alternately to sources of suction and pressure. The valve shall be placed in a horizontal position and a suction of 6 inches Hg absolute applied, and the source of suction cut off. The level of the manometer shall not change more than 1 inch within a period of 1 minute due to leaks in the valve.
The Leak test shall be repeated with a pressure of 105 inches Hg absolute, and the level of the manometer shall not change within a period of 1 minute due to leaks in the valve.

4.4.2 Sampling tests. Five valves shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests. A lot shall consist of not more than 500 valves manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.2.1 Deterioration. The valve shall be submerged in aromatic fuel for a period of 24 hours at a temperature of from 20° to 25°C. The valve shall then be submerged in tap water for the same period of time and at the same temperature. After submerging in aromatic fuel and tap water, the valve shall be subjected to the leak test specified herein, and the same results shall apply.

4.4.2.2 Temperature test. The valve shall be subjected to a temperature of from -54° ±2°C for a period of 4 hours and then subjected to the leak test specified herein. The valve shall then be subjected to a temperature of +71° ±2°C for a period of 4 hours and subjected to the leak test. The same results shall be obtained for both conditions of temperature as specified for the leak test under individual tests.

4.4.3 Special tests. Two valves representing the manufacturer's contract or order shall be subjected to any or all of the following tests.

4.4.3.1 Durability test. The valves shall be actuated for 2,000 times and then subjected to the leak test specified herein. The same results shall be obtained as specified in the individual tests, and there shall be not evidence of wear.

4.4.3.2 Vibration test. The valves shall be subjected to vibration tests, Method 514.3 of MIL-STD-810C. The valves shall be subjected to and meet the requirements of the leak test during the vibration period. At the conclusion of this test, the valves shall be examined. There shall be no damage or looseness.

4.4.3.3 Low temperature exposure. The valve shall be placed within a test chamber and the chamber cooled to and maintained at a temperature of -62°C (-80°F) for a period of 72 hours at which time the valve shall be subjected to and meet the requirements of the leak test specified herein. The temperature of chamber shall then be raised to -54°C (-65°F and maintained for an additional 24-hour period or until stabilization of the valve temperature is reached (Temperature stabilization of equipment may be checked by thermal measuring device in good thermal contact with largest, centrally located internal mass.), which is longer. At the conclusion of this exposure period and while at this temperature, the valve shall be inspected, subject to and meet requirements of the leak test specified herein. The valve temperature shall then be returned to room temperature (68° to 77°F). The valve shall be operated, inspected visually for any deterioration, subjected to and meet leak test requirements specified herein. No damage to the valve shall result from this test.

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4.4.3.4 Magnetic property. At the discretion of the Inspector, the magnetic property of the valve shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long having a magnetic field with a horizontal intensity of 0.17 to 0.19 gauge, when the valve is held in various positions on an east-west line 5 inches from the center of the magnet. (An aircraft compass with compensating magnets removed therefrom may be used as the free magnet for this test.) The maximum deflection of the magnet shall not exceed 1 degree.

4.4.4 Rejection and retest. When tests are specified on a quantity of valves that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, additional valves of the lot represented shall be tested immediately to determine the extent of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the valves.

4.5 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL- STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C, or Commercial in accordance with IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Valve shall be cleaned IAW process C-1 of MIL-P-116.

5.1.1.2 Drying. Valve shall be dried IAW process D-1 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each valve shall be packaged in quantity unit packs of one each IAW Method IC-3 of MIL-P-116. Each valve shall be placed in a PPP-B-636 fiberboard container, weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmissions does not exceed peak values in G's established for the valve when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each valve shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

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5.1.3 Commercial. The commercial preservation of the valve shall be IAW ASTM D3951.

5.2 Packing. Packing shall be Level A, B, C, or commercial as specified (see 6.2).

5.2.1 Level A. Valve packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare, and cube consistent with the protection required.

5.2.2 Level B. Valve packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, Class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Commercial. The packaged valve shall be packed IAW ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve covered by this specification is intended for use in aircraft to purge the line between the manifold pressure gage and engine of condensate and foreign matter.

6.1.1 Operation. This operation is accomplished by opening the valve during engine idling conditions thus sucking air through the line from the cabin or cockpit area to the engine.

6.1.2 Use. The use of the valve is required on all direct reading manifold pressure gage lines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. When first article is required (see 3.1).
d. Level of preservation and packing (see 5.1 and 5.2).

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6.3 First article. When the first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1); and the number of items to be tested as specified in 4.2.

6.4 Subject term (key word) listing.

Condensate
Drain Cock
Engine
Pneumatic

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
NAVY -AS
AIR FORCE -99

Preparing Activity:
AIR FORCE -71

Project Number:
1650-0500

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-V-6618C	2. DOCUMENT DATE (YYMMDD) 8 JANUARY 1993

3. DOCUMENT TITLE
VALVE, STOP CHECK, PURGE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME OC-ALC/TICLA ATTN: HARLENA Y. EDWARDS	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON DSN 336-5960
c. ADDRESS (Include Zip Code) TINKER AFB OK 73145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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